

Adaptive localization by observation assignment for the modified-Cholesky ensemble Kalman filter

Working draft

September 14, 2026

Abstract

Localization in ensemble data assimilation is normally parameterized by a radius of influence, tuned once against a known truth and then held fixed. We replace the radius by a different object: each model component is updated by exactly one observation, chosen from the few that its neighbourhood reaches, and the radius is whatever distance that choice implies. The choice is made by minimizing a variational cost evaluated at a scalar local analysis, computed from ensemble anomalies and observations alone, with no reference trajectory and nothing withheld. Because the cost decomposes over components, the optimal assignment is obtained exactly, in closed form, at a cost of microseconds per evaluation. The resulting radius field adapts to the local observation density without any parameter controlling it, and the observation selected is very often not the nearest one: on a quasi-geostrophic testbed the second, third and fourth candidates are chosen about as often as the first, because what decides the choice is the ensemble correlation and not the distance. The assignment determines the predecessor sets of a modified-Cholesky precision matrix, which is assembled once per cycle and used for a single global analysis.

1 The parameter nobody can pin down

The analysis step of an ensemble Kalman filter combines a forecast ensemble with observations. With an ensemble far smaller than the state dimension, the sample covariance is rank deficient and carries spurious correlations between distant variables; localization removes them. Every localization scheme depends on a radius of influence, and the radius is normally chosen by sweeping it against a known truth in a synthetic experiment and then freezing it.

Attempts to estimate the radius instead run into a difficulty that is not a matter of tuning. A radius per model component is a free parameter with nothing in the data to pin it down. In experiments preceding this work, on a system of 40 components, an oracle with full access to the truth reduced the analysis error by 59% on the cycle it optimized, and the same radius field made the error 22% *worse* on the following cycle, performing worse than a random shuffle of its own values. What such a search finds is the cycle's noise.

Grouping components reduces the count but introduces a grouping that must itself be justified, and the radius within a group remains a free number.

2 Assignment instead of radius

We propose a different parameterization. Let the model state have components $i = 1, \dots, n$ and let there be m observations at known locations. For each component, grow a neighbourhood until it contains κ observations; call them the *candidates* of i and write $\mathcal{C}(i)$ for the set. The component is then updated by exactly one member of $\mathcal{C}(i)$, or by none at all.

The decision variable is the assignment

$$a : \{1, \dots, n\} \longrightarrow \mathcal{C}(i) \cup \{\emptyset\}, \tag{1}$$

and the radius of component i is not chosen: it is the distance to $a(i)$.

Three properties follow immediately.

The radius adapts to the observation density with no parameter. Where observations are dense the neighbourhood stops growing early; where they are sparse it grows further. On the testbed below, a regular lattice of spacing four gives a mean radius of 2.7 and a maximum of 4, while a random network of the same overall density gives a mean of 1.7 and a maximum of 5 — the second number is larger precisely because a random network leaves empty regions. No setting controls this.

The nearest observation is not automatically the right one. What determines how much an observation tells us about a component is the ensemble correlation between them, not the distance. In a flow with structure, two points aligned along a current can be far better correlated than two adjacent points across a front. This is why the neighbourhood keeps growing past the first observation: the choice is made on the cost, and the cost sees the correlation.

Abstaining is a legal choice. If none of the candidates correlates with the component, updating it from any of them injects noise, which is what localization exists to prevent. Leaving the component at its background value is one of the options, and the cost selects it on its own.

3 The cost

For component i assigned to observation j , the local analysis is the scalar Kalman update

$$x_i^a = x_i^b + \frac{\sigma_{ij}}{\sigma_{jj} + r_j}(y_j - x_j^b), \quad (2)$$

with σ_{ij} the ensemble covariance between component i and the observed site of j , σ_{jj} the ensemble variance there, and r_j the observation error variance. The cost is the variational one evaluated at that analysis,

$$J_i(a) = \underbrace{\frac{(x_i^a - x_i^b)^2}{\sigma_{ii}}}_{\text{departure from the background}} + \underbrace{\frac{(y_j - x_j^a)^2}{r_j}}_{\text{residual}}, \quad j = a(i), \quad (3)$$

and $J_i = (y_j - x_j^b)^2/r_j$ when i abstains, since the background is kept and the departure term vanishes. The global score is

$$J(a) = \frac{1}{n} \sum_{i=1}^n J_i(a). \quad (4)$$

No reference trajectory enters (3), and no observation is withheld. Everything is a dot product over the N ensemble members. An assignment that moves the state a long way without justification pays in the first term; one that leaves a residual it could have removed pays in the second.

Proposition 1 (Exact minimization). *J decomposes over components, so $\arg \min_a J(a)$ is obtained by minimizing each J_i independently over $\mathcal{C}(i) \cup \{\emptyset\}$.*

Remark 1. *This is worth stating plainly because it removes a whole apparatus. There is no combinatorial search here: the optimal assignment is a per-component $\arg \min$ over $\kappa + 1$ pre-computed numbers. Metaheuristics applied to (4) return exactly the same assignment — in the experiments below, tabu search at 2×10^5 evaluations and simulated annealing at 1.6×10^4 both reproduce the greedy solution to every digit. The decomposition is a feature of this cost, not of the implementation, and it is lost as soon as a term coupling neighbouring components is introduced.*

Algorithm 1 One assimilation cycle

- 1: propagate the ensemble; form anomalies $\Delta\mathbf{X}$
 - 2: **for each component** grow its neighbourhood until it holds κ observations
 - 3: precompute σ_{ii} , σ_{jj} and σ_{ij} for every candidate pair
 - 4: **for each component** take $a(i) = \arg \min J_i$ over $\mathcal{C}(i) \cup \{\emptyset\}$
 - 5: read off the radius of every component from a
 - 6: assemble $\hat{\mathbf{B}}^{-1}$ once and solve for the analysis
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4 From assignment to analysis

The assignment determines a radius per component, and therefore the predecessor sets of the modified-Cholesky estimator [1, 2]. For each component i with predecessors $P(i)$ given by its radius,

$$\beta_i = (\mathbf{X}_P^\top \mathbf{X}_P + \alpha \mathbf{I})^{-1} \mathbf{X}_P^\top \Delta \mathbf{X}_i, \quad \mathbf{L}_{ii} = 1, \quad \mathbf{L}_{i,P(i)} = -\beta_i^\top, \quad (5)$$

with $\mathbf{D}_{ii}$ the inverse residual variance, and $\hat{\mathbf{B}}^{-1} = \mathbf{L}^\top \mathbf{D} \mathbf{L}$. The analysis solves

$$(\hat{\mathbf{B}}^{-1} + \mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H}) \delta \mathbf{x} = \mathbf{H}^\top \mathbf{R}^{-1} (\mathbf{y} - \mathbf{H} \mathbf{x}^b). \quad (6)$$

The precision is positive definite for any assignment, because it retains the form $\mathbf{L}^\top \mathbf{D} \mathbf{L}$; truncating an already-formed precision does not preserve this, and in a companion measurement produced an indefinite matrix in 44 of 44 cases tested.

5 Numerical experiments

The testbed is the 1.5-layer quasi-geostrophic model of Sakov and Oke [4],

$$q_t = -\psi_x - \varepsilon J(\psi, q) - A \Delta^3 \psi + 2\pi \sin(2\pi y), \quad q = \Delta \psi - F \psi, \quad (7)$$

on the unit square with $F = 1600$, $\varepsilon = 10^{-5}$ and Dirichlet boundaries, at 49×49 per field. The integration is carried on q and ψ is recovered from it at every step, so ψ is diagnostic and only q is estimated: $n = 2401$. Ensemble members and the truth are drawn from snapshots of a long run, following [4]. The dissipation is ten times the model default, which is what gives a stationary regime: at the default the norm of q grows by a factor of fourteen between $t = 500$ and $t = 8000$ and there is no climatology to sample.

5.1 What the assignment does

With 144 observations on a lattice of spacing four and $\kappa = 4$:

Assignment	J
Every component abstains	123.82
Every component takes its nearest observation	0.587
Optimal assignment	0.052
Random assignment	30.28

Table 1: The cost separates good assignments from bad ones by orders of magnitude, and the optimal assignment improves on taking the nearest observation by a factor of eleven.

The distribution of choices is the result worth emphasising. Of the 2401 components, 516 take their first candidate, 603 their second, 647 their third and 635 their fourth. The candidates are ordered by distance, so the nearest observation is chosen barely a fifth of the time: the ensemble correlation, not the distance, is what decides.

5.2 Cost

Evaluating (4) takes 1.6×10^{-2} ms. Building the candidate sets and precomputing the covariances is a fraction of a second per cycle and is done once. For comparison, one evaluation of an objective that requires forming the precision and solving the analysis system costs 0.18 s at the same resolution, four orders of magnitude more; a radius large enough to destroy the sparsity of the precision — 51% non-zero at radius 12 against 0.5% at radius 1 — pushes that to 1.07 s.

6 Discussion

The contribution is a reparameterization. The radius stops being a free number and becomes a consequence of a choice among observations that are actually present, which is what makes it estimable: a choice among $\kappa + 1$ alternatives is constrained in a way that a positive real number is not.

The honest limitation is stated in the remark above. This cost decomposes, so the optimization is trivial and no search is required. That is a virtue operationally and a limitation scientifically: it means the assignment formulation, as posed here, does not exercise the combinatorial structure it appears to have. Coupling neighbouring components — penalizing roughness in the analysis increment, or capping how many components a single observation may update — breaks the decomposition and makes the problem genuinely combinatorial. Whether that coupling improves the analysis, rather than merely making the optimization harder, is a separate question and is left to subsequent work.

References

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